

“Warning”

□ ESP32-S3-RS485-CAN Industrial Development Board Dual Core 240MHz Wi-Fi Bluetooth 5 LE Isolated RS485 CAN Bus DIN Rail Case DC7-36V Wide Voltage With RTC Clock For Modbus IoT Automation Sensor Monitoring Industrial grade ESP32-S3 controller with fully isolated RS485 and CAN bus, DIN rail mounting enclosure, wide 7-36V industrial power input, built-in RTC and status indicator lights, perfect for Modbus sensor data acquisition and IoT automation projects.

5 Core Selling Points

1. Powerful ESP32-S3 Dual Core Industrial Main Controller Equipped with Xtensa 32-bit LX7 dual-core ESP32-S3R8 chip, main frequency up to 240MHz, strong AI computing performance to handle complex industrial data processing, control logic and IoT transmission tasks stably.
2. Stable Dual-Mode Wi-Fi + Bluetooth 5 LE Wireless Communication Integrated 2.4GHz 802.11 b/g/n Wi-Fi and low-power Bluetooth 5 LE dual wireless module, excellent RF anti-interference performance, support remote wireless data upload, device pairing and IoT gateway functions.
3. Full Galvanic Isolated RS485 & CAN Bus For Safe Industrial Communication Adopt complete digital isolation and integrated power isolation design for RS485 and CAN interfaces, effectively block external high voltage signal interference to protect main chip; built-in TVS surge & ESD protection for industrial field harsh environment.
4. Dedicated Status Indicator Lights For Real-Time Operation Monitoring Independent power indicator, RS485 transmit/receive dual color lights and CAN bus communication flash light, users can directly judge data transmission status without extra detection tools, convenient on-site equipment debugging.
5. DIN Rail Protective Enclosure For Easy Industrial Installation Integrated DIN rail plastic housing, can be directly clipped on standard electrical cabinet guide rail, fully wrap internal circuit for dust and collision protection, match standard industrial cabinet wiring installation standard.

Full Technical Specifications

Item Parameter Model ESP32-S3-RS485-CAN Industrial Dev Board Main Chip ESP32-S3R8 Dual Core LX7 240MHz Wireless 2.4G Wi-Fi 802.11 b/g/n + Bluetooth 5 LE Communication Port Isolated RS485 + Isolated CAN Bus Terminal Power Input DC 7V ~ 36V Wide Industrial Voltage Download Interface USB Type-C (Power + Debug + Firmware Flash) Extra Feature Onboard RTC Clock, Hardware Full Isolation, Surge & ESD Protection Mounting Method Standard DIN Rail Protective Enclosure Dimension 91.6 * 23.3 * 58.7 mm

Application Scenarios

- Industrial Modbus RS485 Sensor Data Monitoring & Acquisition
- CAN Bus Equipment, New Energy Equipment Signal Collection
- Factory PLC Automation Retrofit & IoT Wireless Gateway
- Smart Agriculture Environmental Sensor Control System
- DIY Industrial Controller, Equipment Remote Monitoring Project

□ Complete Package List

- 1 x ESP32-S3-RS485-CAN DIN Rail Industrial Module
- 1 x 150mm SH1.0 4PIN Connecting Cable
- 1 x Matching Small Screwdriver

23.3mm/0.91in



91.7mm/3.61in



58.8mm/2.31in



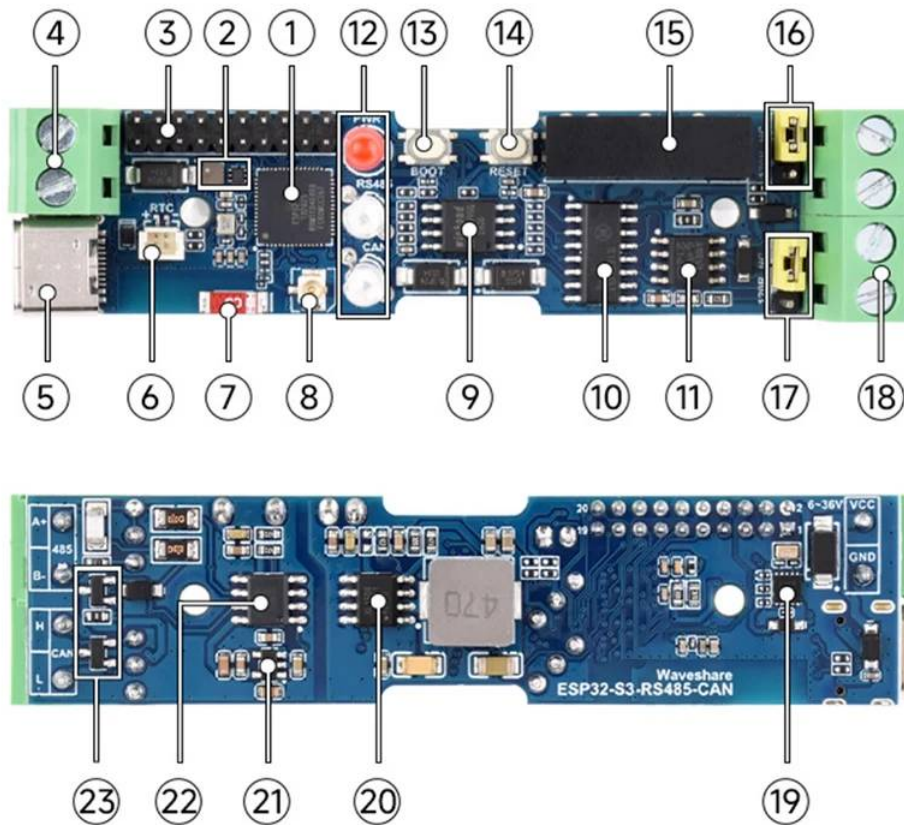
47.3mm/1.86in



Product Specifications

Model: ESP32-S3-RS485-CAN	automatically detects and controls data transmission direction
Microcontroller: ESP32-S3 (default chip: ESP32-S3R8)	Interface Protection: Provides TVS diode protection, surge and ESD protection
Wireless Communication: 2.4GHz Wi-Fi (802.11 b/g/n), Bluetooth 5 (LE)	Matching Resistor: Default NC, reserved for 120Ω resistor, enabled via jumper cap
USB Interface Type: USB Type-C	PWR Power Indicator: Connects to USB; illuminates red when voltage detected
Supply Voltage: 5V	RS485 Indicators:
Functions: Device power supply, USB communication, firmware download, etc.	Illuminates green when data is transmitted from the RS485 interface
Isolated RS485 Communication Interface:	Illuminates blue when data is received back via the RS485 interface
Interface Type: Terminal Block	CAN Indicator: Flashes when data communication occurs on the CAN interface
Direction Control: Automatically controlled via host hardware flow setting	Terminal Supply Voltage Range: 7V ~ 36V
Interface Protection: Provides TVS diode protection, surge and electrostatic protection	Enclosure: Rail-mounted protective enclosure
Matching Resistor: Default NC, reserved for 120Ω resistor, enabled via jumper cap	
Isolated CAN Communication Interface:	
Interface Type: Terminal Block	
Direction Control: Hardware	

Resource Overview



- 1、ESP32-S3
Operating at up to 240MHz, featuring 2.4GHz Wi-Fi and low-power BT
- 2、DC-DC Power Module
Power module supporting up to 3.3V 2A output
- 3、Pin Header Interface (2.0mm pitch)
Connects to other functional modules via pin headers
- 4、Power Terminal Block
Supports wide voltage input range of DC 7V to 36V
- 5、USB Type-C Port
Supports device power delivery, firmware download, and USB communication
- 6、RTC Battery Connector
Connects to rechargeable RTC battery (SH1.0)
- 7、Surface-mount Ceramic Antenna
- 8、IPEX Gen1 Socket
Can be switched to external antenna by desoldering resistors
- 9、16MB Flash
- 10、Digital Isolation
Prevents external signal interference from affecting the control chip
- 11、CAN Conversion Chip
- 12、Operational Indicator Lights
PWR: Power indicator light
RS485: Green light illuminates when data is transmitted from the RS485 interface;
Blue light illuminates when data is received via the RS485 interface
CAN: CAN indicator light; flashes when data communication occurs on the CAN interface
- 13、Device BOOT Button
- 14、Device Reset Button
- 15、Power Isolation
Provides stable isolated voltage; isolated terminals require no additional power supply
- 16、RS485 Terminating Resistor
Reserved for 120Ω terminating resistor; enabled via jumper cap
- 17、CAN Termination Resistor
Pre-installed 120Ω termination resistor, enabled via jumper cap
- 18、RS485 & CAN Communication Interfaces
Supports connection to external RS485 and CAN devices
- 19、RTC Clock Chip
PCF85063 Real-Time Clock (RTC)
- 20、DC-DC Power Supply Chip
- 21、ME6217C33M5G
Low Dropout LDO, Current (Max) 800mA
- 22、RS485 Conversion Chip
- 23、On-Board TVS (Transient Voltage Suppressor)
Effectively suppresses surge voltages and transient spikes in the circuit

Powerful AI Computing Capabilities

Based on ESP32-S3, dual-core 240MHz, delivering robust AI computing power.



Supports Wi-Fi + BT 5 (LE)

Supports Wi-Fi + BT 5 (LE) for more stable wireless connections



Isolated RS485/CAN Interface

Isolated RS485/CAN Interface, Safer Industrial Communication

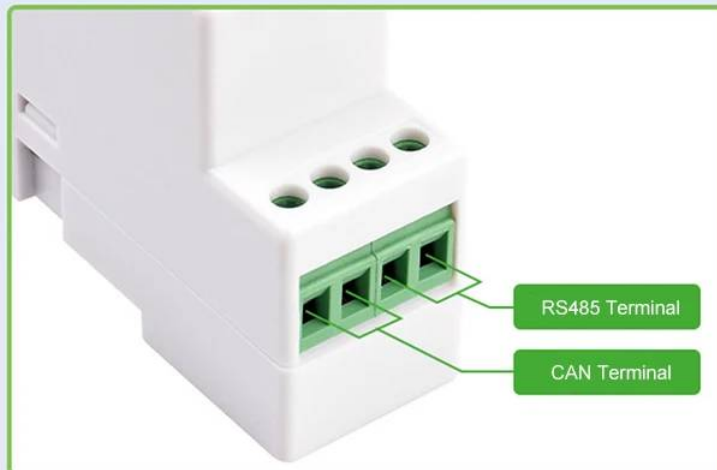
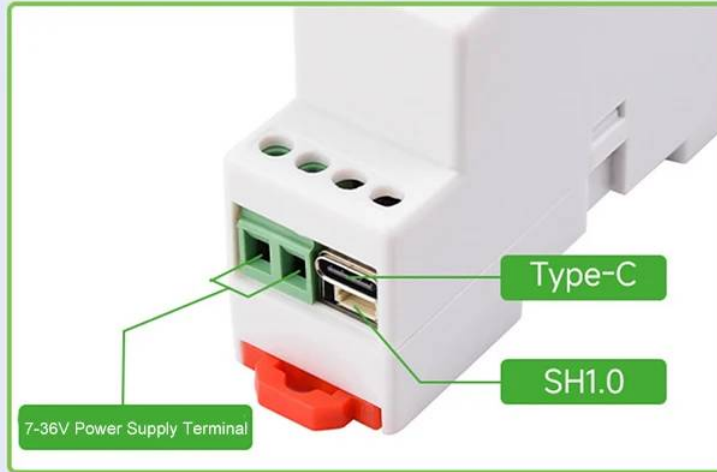


On-board Indicator Lights

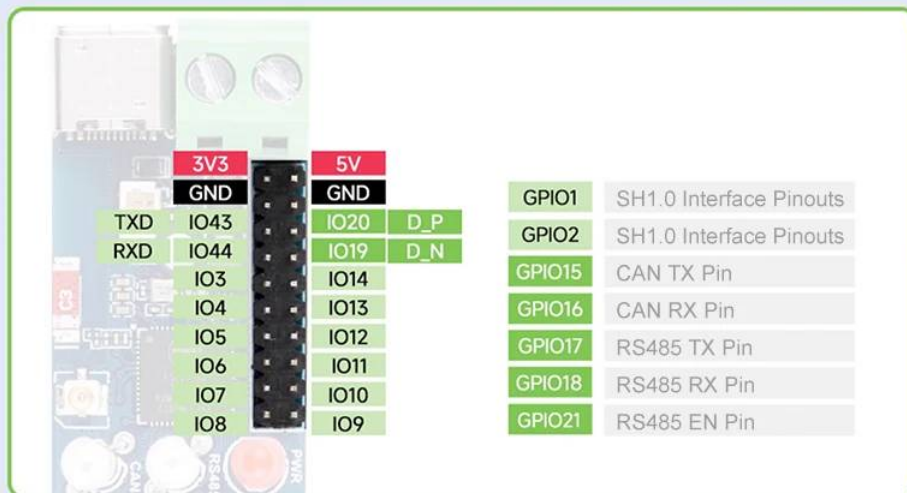
On-board RS485 and CAN indicator lights for easy monitoring of device operational status



Interface Specification

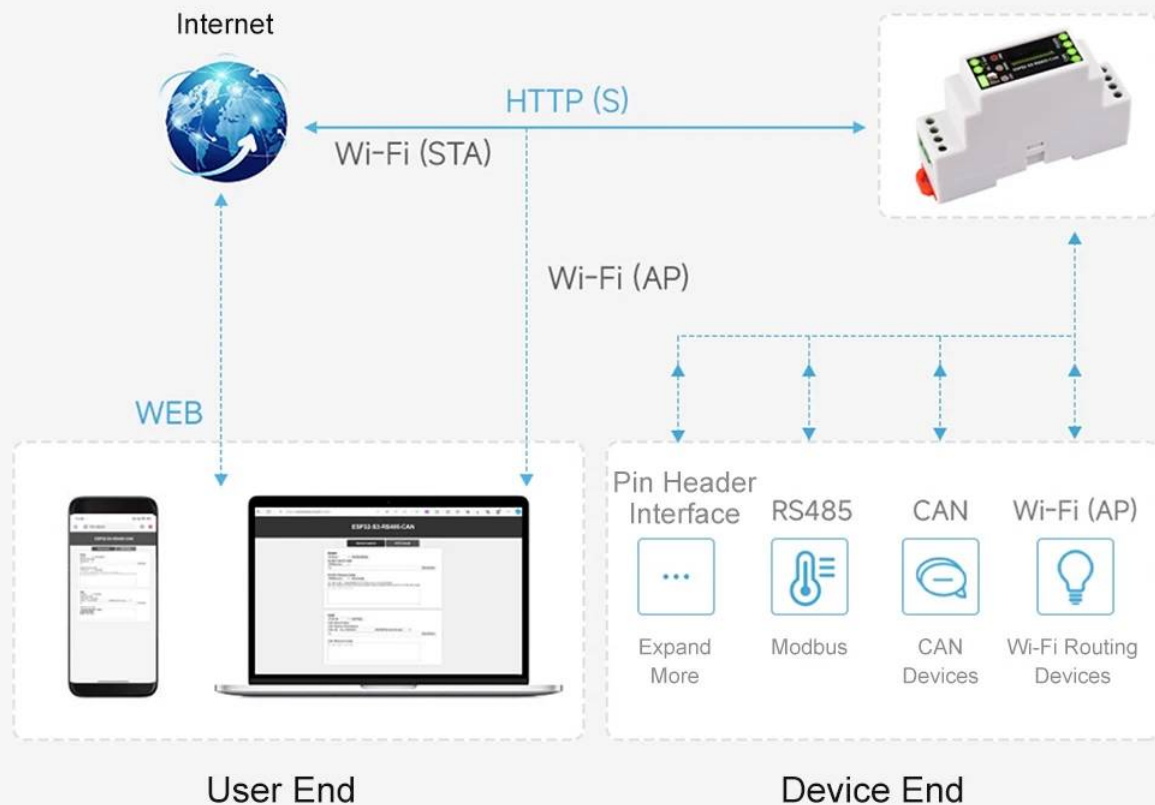


Upper Terminal Description			Lower Terminal Description	
Power supply terminal	Power supply positive terminal: DC 7V to 36V		RS485 Terminal	A+ signal pin
	Negative Power Terminal			
Type-C	Can be used for device power supply, firmware download, and USB communication.			
SH1.0	GND	Ground	CAN Terminal	H signal pin
	3V3	3.3V Power Supply Output		
	GPIO2	Expansion 10		L signal pin
	GPIO1	Expansion IO		



Example Demonstration

The sample code implements a communication interface integrated with a web page to demonstrate IoT connectivity, enabling data visualization services.



Based on ESP32-S3 Designed specifically for the AIoT market

Features an Xtensa 32-bit LX7 dual-core processor Operating at up to 240MHz,
delivering powerful AI processing capabilities and robust security encryption mechanisms
Integrated 2.4GHz Wi-Fi and Bluetooth 5 (LE) dual-mode wireless communication



ESP32-S3 Industrial Control Board Industrial-Grade RS485 and CAN Communicator

Based on the ESP32-S3 microcontroller, supporting Wi-Fi, BT, RS485, and CAN interfaces

Features built-in protection circuits including power isolation and digital isolation for enhanced safety, stability, and reliability



Version Selection



Onboard Antenna Version
ESP32-S3-RS485-CAN



External Antenna Version
ESP32-S3-RS485-CAN-U